VARUNA 2047 — ForestWatch Intelligence Report

Region: WAR | **Type:** District | **Generated:** 2026-10-27 05:38 UTC | **Classification:** Research Grade

1. EXECUTIVE SUMMARY & CANOPY INTEGRITY

This evidence dossier presents monitored canopy changes and satellite observations for WAR. Under the Forest Survey of India (ISFR) benchmark, baseline forest extent is recorded at **38,200 hectares**. During the observation window, measured tree-cover loss area totaled **42.5 hectares** (0.111% of baseline).

2. TREE-LOSS METRICS & DEFENSIBLE ESTIMATION

Metric Indicator	Measured Value	Scientific Standard
Measured Tree-Cover Loss Area	42.5 ha	Hansen GFC v1.11 / Sentinel-2 dNDVI (10m)
Percentage Canopy Change	0.111%	100 * (Loss Area / Baseline Forest Area)
Estimated Tree Stems Lost	~9,350 trees	FSI NFI Average Stem Density per Hectare (+/- 25%)
Change Trajectory	STABLE	Forest loss rate remains within historical baseline bounds (+/- 15%
Baseline Sentinel-2 Scene	S2A_MSIL2A_20190114T050151_N0211_R0...	ESA Copernicus BOA Surface Reflectance
Recent Sentinel-2 Scene	S2B_MSIL2A_20240118T050149_N0510_R0...	ESA Copernicus BOA Surface Reflectance

3. SCIENTIFIC TRANSPARENCY & LIMITATION DISCLOSURES

Important Methodological Note: Tree-cover loss detected: 42.5 hectares (estimated ~9,350 trees, range: 5,950 to 12,750 trees based on FSI inventory density).
Estimated tree count is a biophysical proxy derived from average stem density. Canopy-loss pixels do NOT imply clearcut logging or illegal felling without ground verification.